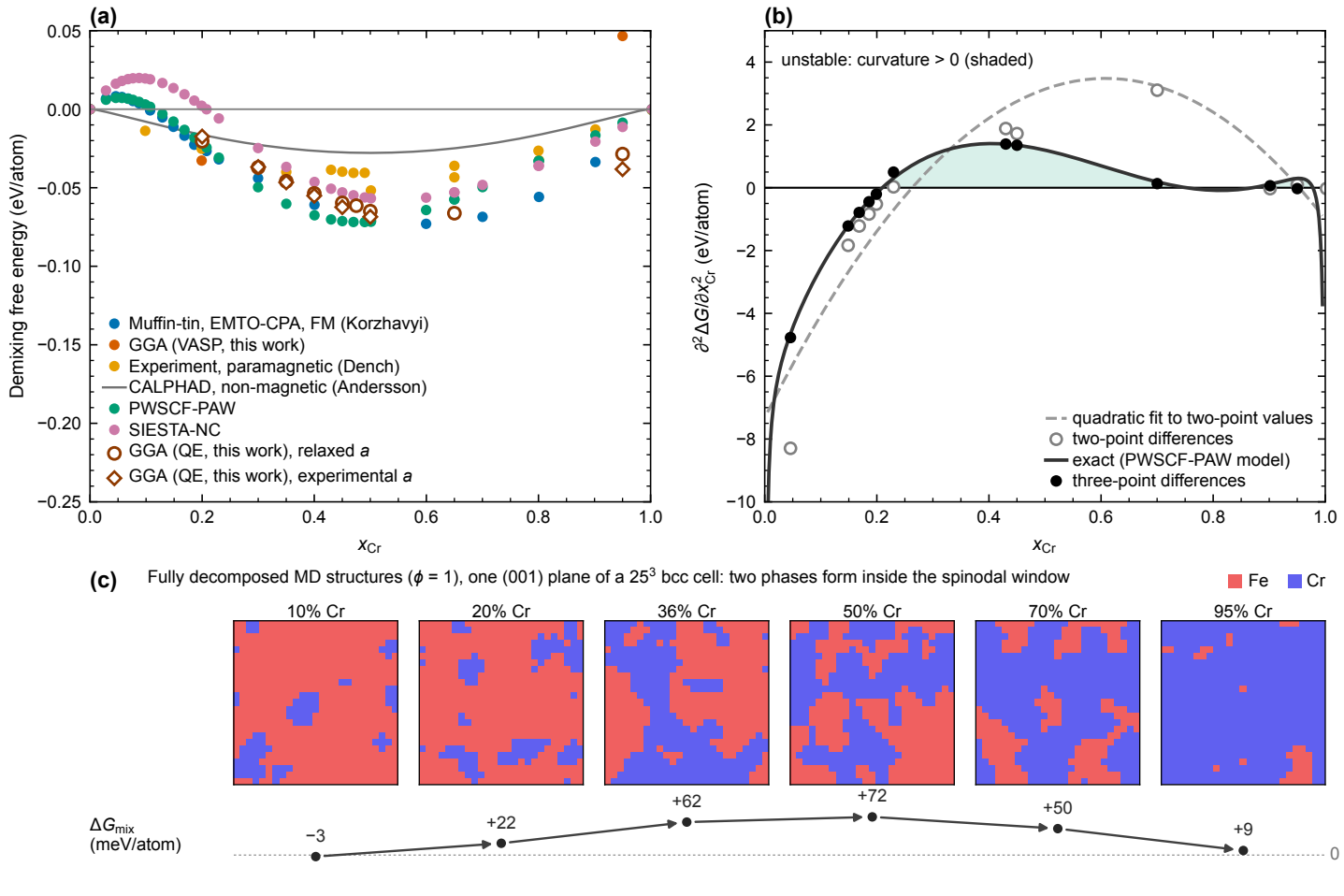

(a)
Demixing free energy (eV/atom)
0.05
0.00
−0.05
−0.10
−0.15
−0.20
−0.25
0.0
0.2
0.4
0.6
0.8
1.0
x_{Cr}
Muffin-tin, EMTO-CPA, FM (Korzhavyi)
GGA (VASP, this work)
Experiment, paramagnetic (Dench)
CALPHAD, non-magnetic (Andersson)
PWSCF-PAW
SIESTA-NC
GGA (QE, this work), relaxed a
GGA (QE, this work), experimental a
(b)
unstable: curvature > 0 (shaded)
$\partial^2 \Delta G / \partial x_{Cr}^2$ (eV/atom)
4
2
0
−2
−4
−6
−8
−10
0.0
0.2
0.4
0.6
0.8
1.0
x_{Cr}
quadratic fit to two-point values
two-point differences
exact (PWSCF-PAW model)
three-point differences
(c)
Fully decomposed MD structures (ϕ = 1), one (001) plane of a 25^3 bcc cell: two phases form inside the spinodal window
Fe
Cr
10% Cr
20% Cr
36% Cr
50% Cr
70% Cr
95% Cr
ΔG_{mix} (meV/atom)
−3
+22
+62
+72
+50
+9
0